

**P Quantities and units in
mechanics**

Name: _____

Class: _____

Date: _____

Time: **9 minutes**

Marks: **8 marks**

Comments:

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Q1.

A particle, of mass 400 grams, is initially at rest at the point O.

The particle starts to move in a straight line so that its velocity, $v \text{ m s}^{-1}$, at time t seconds is given by

$$v = 6t^2 - 12t^3 \text{ for } t > 0$$

- (a) Find an expression, in terms of t , for the force acting on the particle.

(3)

- (b) Find the time when the particle next passes through O.

(5)

(Total 8 marks)



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